Global Convergence and High-Frequency Spectral Bias of Low-Rank Networks

Anonymous Author(s)
Affiliation
Address
email

Abstract

Low-rank neural networks are usually introduced as compressed versions of dense models, but this view misses their main theoretical role: rank creates a structured mean-field system whose optimization and spectral behavior can be analyzed directly. We prove that low-rank random-feature neural networks, with frozen feature maps and trainable channel weights, have well-posed mean-field dynamics and that every convergent trajectory reaches a global minimizer of the population risk under a standard loss-identifiability condition. The result holds for arbitrary depth and standard independent initialization; low rank changes constants through the channel mixing norm but does not destroy the dense-span argument needed for global optimality. We then explain why convergence alone is not enough for highly oscillatory functions. Building on the spectral-bias perspective of Rahaman et al. [8], low rank is shown to control path-space degrees of freedom, continuous piecewise affine geometry, and finite-time Fourier recovery. Experiments on oscillatory targets and Fourier-native rank sweeps support the view that the useful rank is often intermediate: small enough to change spectral transfer, but large enough to avoid an approximation bottleneck.

1 Introduction

Low-rank neural networks replace dense matrices by factored or bottlenecked operators. The standard motivation is computational: fewer trainable parameters, cheaper memory traffic, and sometimes faster training. This paper argues that the more important question is mathematical. Does the low-rank constraint preserve the optimization guarantees available in mean-field theory, and can the same constraint help with the spectral bias that prevents neural networks from learning highly oscillatory functions?

We answer the first question positively for a low-rank random-feature architecture. The model freezes a rich random-feature map and trains low-rank channel weights through mean-field gradient flow. Since the frozen features keep a dense span throughout training, the usual obstruction in deep mean-field convergence proofs is removed: at a limit point, zero gradient implies the conditional first-order condition for the loss, and the loss-identifiability assumption forces global optimality. Low rank enters through the mixing matrix and channel index, changing constants but not the logic of the proof.

The second question is more subtle. A globally convergent training dynamic can still learn low frequencies first and fail at finite time on highly oscillatory targets. We therefore connect the convergence theorem to a rank-dependent spectral bias mechanism. For ReLU networks, low rank constrains active path coefficient tensors; this constrains switching normals, suppresses high-codimension faces in the continuous piecewise affine complex, and changes Fourier shell behavior. In one dimension, the raw spline tail remains close to the classical k^{-4} law, so the correct diagnostic

Submitted to 40th Conference on Neural Information Processing Systems (NeurIPS 2026). Do not distribute.

is not the asymptotic exponent alone but the learned Fourier recovery curve. This leads to a practical rank-selection rule: choose the smallest rank that avoids the approximation bottleneck while flattening the finite-time spectral transfer.

Contributions. The paper makes three contributions. First, we state a global convergence theorem for arbitrary-depth low-rank random-feature networks in the mean-field limit. Second, we turn low-rank spectral bias into a geometric statement: rank constrains active path coefficients, which constrains CPWL switching geometry and Fourier shell energy. Third, we connect this mechanism to experiments on highly oscillatory targets, including high-frequency regression, frozen-vs-trainable random features, and heavy-tail spectral updates.

2 Freezing Some Weights Allows Global Convergence

Let $X \in \mathbb{R}^d$ and Y be drawn from a population distribution. The finite-width low-rank random-feature network used throughout the paper is the one from the original experiments:

$$h^{(0)}(x) = x, \qquad h^{(\ell)}(x) = \frac{1}{n_\ell} \sum_{j=1}^{n_\ell} w_j^{(\ell)} \varphi_\ell\Big(\langle L_j^{(\ell)}, h^{(\ell-1)}(x) \rangle + b_j^{(\ell)} \Big), \tag{1}$$

where the feature directions $L_j^{(\ell)}$ and biases $b_j^{(\ell)}$ are frozen, while the channel weights $w_j^{(\ell)}$ are trained. This is a low-rank factorization of a dense layer in which the left factor is fixed: if $W_\ell = U_\ell V_\ell^\top$ with $U_\ell \in \mathrm{St}(n_\ell, r_\ell)$, then U_ℓ plays the role of an orthogonal random-feature or mixing basis and the trainable factor V_ℓ is absorbed into the channel weights. Thus one may avoid training U_ℓ entirely, either by drawing it once as a random orthogonal matrix or by fixing it on the Stiefel manifold; training only the right/channel factor is the regime analyzed below.

In the mean-field limit, frozen feature maps are written $L^0(C_\ell)$ and trainable channel weights are written $w_\ell(t, C_\ell)$. The channel mixer $L \in \mathbb{R}^{N_2 \times r}$ is taken to have orthonormal columns, $L^\top L = I_r$. It is the left orthogonal factor of a rank-r factorization of the second-layer weight matrix: if a dense second-layer operator has truncated singular-value decomposition $W_2 = U_r \Sigma_r V_r^\top$, we set $L = U_r$ and absorb $\Sigma_r V_r^\top$ into the rank-channel functions below. In the two-hidden-layer case, the first partial functions are

$$f_k(t, x) = \mathbb{E}_{C_1}\left[w_1(t, C_1, k)\, \varphi_1\left(\langle L^0(C_1), x \rangle \right) \right], \qquad k = 1, \ldots, r, \tag{2}$$

and the second-layer preactivation is

$$H_2(t, c_2; x) = \sum_{k=1}^{r} L_{c_2,k} f_k(t, x). \tag{3}$$

Thus H_2 is the reconstruction of the second-layer signal from an orthogonal rank-r channel basis. The output is obtained by averaging the trainable top weights against $\varphi_2(H_2)$. For deeper networks, the same alternating pattern is used: trainable channel weights are averaged against frozen random features and mixed through bounded low-rank channel matrices.

The population objective is

$$\mathcal{L}(W) = \mathbb{E}_{(X,Y)}\left[\mathcal{L}\left(Y, \hat{y}(X; W) \right) \right]. \tag{4}$$

The mean-field gradient flow evolves the trainable weights by

$$\partial_t w_\ell(t, \cdot) = -\xi_\ell(t)\, \nabla_{w_\ell} \mathcal{L}(W(t)), \tag{5}$$

where ξ_ℓ are bounded learning-rate schedules. The exact formulas are standard backpropagation in the mean-field variables; the only low-rank modification is the channel mixing through $L_{c_\ell,k}$.

Assumption 2.1 (Regularity and diversity)**.** *There is $K \geq 1$ such that the activations are K-Lipschitz, the relevant activation derivatives are bounded, the loss derivative $\partial_2 \mathcal{L}$ is bounded and Lipschitz, and the orthogonal channel mixers satisfy $L^\top L = I_r$ together with $\sup_{c_\ell} \sum_{k=1}^{r} |L_{c_\ell,k}| \leq rK$. Inputs and frozen features are bounded. The support of the first frozen feature law is dense in $\mathbb{R}^d$, and the activation is non-polynomial, so the frozen random features have dense span in $L^2(\mathbb{P}_X)$.*

Assumption 2.2 (Loss identifiability and convergence)**.** *The loss is non-negative and satisfies the implication $\partial_2 \mathcal{L}(y, u) = 0 \Rightarrow \mathcal{L}(y, u) = 0$ in the conditional population sense. The mean-field trajectory has a limit point in the natural Wasserstein-Orlicz topology used by the gradient-flow system.*

3 Global Convergence

The first result is well-posedness. It ensures that the low-rank channel formulation defines a genuine limiting dynamic rather than only a formal ODE.

Theorem 3.1 (Well-posedness of low-rank mean-field ODEs)**.** *Under Assumption 2.1 and sub-Gaussian initialization of the trainable weights, the low-rank mean-field gradient-flow system has a unique solution on every finite interval* $[0, T]$ *and therefore on* $[0, \infty)$.

We use this theorem only to justify that the limiting dynamics are well-defined. The technical fixed-point estimates are standard mean-field estimates with the dense-layer constants replaced by low-rank mixing norms, and are not the focus of the paper.

Theorem 3.2 (Global convergence of RF-LR training)**.** *Under Assumptions 2.1–2.2, if the low-rank mean-field dynamics converges to* $W^\star$*, then* $W^\star$ *is a global minimizer of the population loss:*

$$\mathcal{L}(W^\star) = \inf_{W} \mathcal{L}(W). \tag{6}$$

For every depth $L \geq 2$*, this holds with standard independent initialization of the trainable weights.*

Proof idea. At a limit point, the gradient-flow velocity is zero. Testing the top-layer equation against arbitrary frozen features yields orthogonality between the loss derivative and the closed span of the represented features. Because the frozen feature law has full support and the activation is non-polynomial, this span is dense in $L^2(\mathbb{P}_X)$. Hence

$$\mathbb{E}\left[\partial_2 \mathcal{L}\left(Y, \hat{y}(X; W^\star)\right) \mid X = x\right] = 0 \quad \text{for } \mathbb{P}_X\text{-a.e. } x. \tag{7}$$

The loss-identifiability assumption then implies zero conditional loss and therefore global optimality. In dense deep mean-field proofs, one must show that the trained first-layer support remains rich. Here the support is frozen, so the dense-span property is automatic along the whole trajectory. Low rank only changes the upstream factors in the backpropagated signal. □

4 Low-Rank Spectral Bias

Related work: spectral bias. Theorem 3.2 says that any convergent trajectory reaches a global minimizer. It does not say how fast different frequencies are learned, nor which finite-time solution is reached under a fixed training budget. Highly oscillatory functions expose this gap. This is precisely the phenomenon called spectral bias by Rahaman et al. [8]: standard neural networks tend to fit low-frequency components before high-frequency components. Rahaman et al. [8] explain this behavior through Fourier decay and training dynamics of neural networks.

Our mechanism is complementary and more structural. In ReLU networks, the represented function is continuous piecewise affine (CPWL), so Fourier behavior can be related to the polyhedral complex induced by activation boundaries. Low rank does not merely reduce parameter count; it constrains the algebraic degrees of freedom of the active path coefficients, and this changes which CPWL faces can carry Fourier mass. Our goal is therefore not to contradict classical spectral bias, but to identify rank as an architectural knob that modifies the geometry behind the finite-time Fourier transfer.

In a ReLU network, on every activation region the network is affine, and the affine coefficient is a sum over active paths. The high-level idea is that a low-rank network has fewer *independent* paths in this sum-over-paths representation. More precisely, when $W_\ell = U_\ell V_\ell^\top$ with rank r_ℓ, the path coefficients involving layer ℓ must factor through an r_ℓ-dimensional latent coordinate. Thus the graph may still contain many symbolic paths, but the number of independent path constraints is much smaller than in a full-width layer. Fewer independent path constraints mean fewer independent switching constraints for the polytopes of the CPWL linear-region complex. Geometrically, this favors more high-dimensional faces and suppresses low-dimensional, high-codimension intersections. Spectrally, the Fourier shell law below then predicts a lower finite-resolution decay, closer to the facet-dominated regime.

This algebraic constraint becomes geometric. Inside a cell of the previous layers, a new switching normal has the form

$$n_{\ell i \varepsilon} = B_{\ell-1,\varepsilon}^\top V_\ell u_{\ell i} \in \operatorname{Im}(B_{\ell-1,\varepsilon}^\top V_\ell), \qquad \dim \operatorname{Im}(B_{\ell-1,\varepsilon}^\top V_\ell) \leq \min(d, r_1, \ldots, r_\ell). \tag{8}$$

Thus rank lowers the dimension of the normal subspace and suppresses high-codimension faces first.

Face Fourier expansion. Let $g = \chi f$, where f is CPWL on a finite polyhedral complex in $\mathbb{R}^d$ and $\chi \in C_c^\infty(\mathbb{R}^d)$ is an observation window. Let $\mathcal{F}_q(f)$ be the codimension-q faces of the complex. For every large frequency $\xi = \rho\theta$, $\theta \in \mathbb{S}^{d-1}$, the Fourier transform has the expansion

$$\widehat{g}(\rho\theta) = \sum_{q=1}^{d} \rho^{-(q+1)} \sum_{F \in \mathcal{F}_q(f)} A_F(\rho, \theta) + O(\rho^{-(d+2)}), \tag{9}$$

where A_F is a bounded oscillatory face coefficient controlled by the jump Δ_F of the appropriate normal derivative across the local fan around F. In particular,

$$|A_F(\rho, \theta)| \leq C_\chi \|\Delta_F\| \operatorname{vol}_{d-q}(F)\, \omega_F(\theta). \tag{10}$$

This is the anisotropic phenomenon isolated by Rahaman et al. [8]: in almost all directions a ReLU network has fast $k^{-(d+1)}$ amplitude decay, while in special directions orthogonal to facets of the linear regions the decay can be as slow as k^{-2}. Thus low codimension faces are precisely the geometric structures that keep high-frequency Fourier mass visible at finite resolution.

Proposition 4.1 (Fourier shell law for CPWL networks)**.** *Let $g = \chi f$ be a windowed continuous piecewise affine network in dimension d. Define the codimension-q face moment*

$$M_q(f) = \sum_{F \in \mathcal{F}_q(f)} \|\Delta_F\|^2 \operatorname{vol}_{d-q}(F)^2 \mathbb{E}_{\theta \in \mathbb{S}^{d-1}} \omega_F(\theta)^2. \tag{11}$$

Then the shell-averaged Fourier energy satisfies

$$S_f(\rho) = \mathbb{E}_{\|\xi\| \simeq \rho} |\widehat{g}(\xi)|^2 \lesssim \sum_{q=1}^{d} M_q(f) \rho^{-2(q+1)}. \tag{12}$$

Low rank suppresses M_q for larger q first, moving finite-resolution spectra toward the facet-dominated ρ^{-4} energy regime. This should be read as a coefficient statement rather than a claim that low rank creates a new universal exponent: the useful comparison with full rank is the reduction of M_q/M_1 and of the transition coefficient $(M_q/M_1)\rho^{-2q+2}$ for $q \geq 2$.

Proof. Write the CPWL network on its polyhedral complex as

$$f(x) = \sum_{\varepsilon} (a_\varepsilon^\top x + b_\varepsilon) \mathbf{1}_{P_\varepsilon}(x), \qquad g(x) = \chi(x) f(x). \tag{13}$$

The cutoff makes g compactly supported, so its Fourier transform is an ordinary oscillatory integral. On each cell P_ε, integrate by parts in the direction $\theta = \xi/\|\xi\|$. The affine interior terms either gain powers of ρ^{-1} or cancel with the adjacent cell contribution; the non-canceling terms are the jumps of normal derivatives across the faces of the polyhedral complex.

Iterating this argument localizes the qth boundary contribution on codimension-q faces. A codimension-q face appears after q boundary reductions plus one affine derivative, which gives the factor $\rho^{-(q+1)}$. The remaining integral is over the face itself and has the form

$$A_F(\rho, \theta) = \int_F e^{-i\rho\langle \theta, x \rangle} B_F(\theta, x)\, d\sigma_F(x), \tag{14}$$

where B_F is linear in the jump Δ_F of the appropriate normal derivative and in derivatives of the window χ. Hence

$$|A_F(\rho, \theta)| \leq C_\chi \|\Delta_F\| \operatorname{vol}_{d-q}(F) \omega_F(\theta). \tag{15}$$

This is the same anisotropic mechanism as the spectral-decay formula of Rahaman et al. [8]: generic directions have fast $k^{-(d+1)}$ amplitude decay, but directions orthogonal to facets can decay as slowly as k^{-2}.

Squaring the face expansion gives diagonal face terms and oscillatory cross terms. The diagonal contribution of codimension-q faces is bounded by

$$\rho^{-2(q+1)} \sum_{F \in \mathcal{F}_q(f)} \|\Delta_F\|^2 \operatorname{vol}_{d-q}(F)^2 \omega_F(\theta)^2. \tag{16}$$

The off-diagonal terms are oscillatory products between distinct faces. After averaging over a shell, they are either lower order by non-stationary phase or bounded by Cauchy–Schwarz and absorbed into the same face moments. Taking the angular expectation yields

$$S_f(\rho) \lesssim \sum_{q=1}^{d} M_q(f)\rho^{-2(q+1)}. \tag{17}$$

Thus the exponent $2(q+1)$ is inherited from face codimension, while the rank-dependent quantities are the coefficients $M_q(f)$. Low rank acts on these coefficients by reducing the number and strength of high-codimension intersections. □

In one dimension, this mechanism has a crucial caveat. A ReLU network is a linear spline, and for derivative jumps Δ_j at knots t_j,

$$\widehat{f}(k) = -\frac{1}{(ik)^2}\sum_j \Delta_j e^{-ikt_j} + O(|k|^{-3}), \qquad |\widehat{f}(k)|^2 \sim |k|^{-4}A(k). \tag{18}$$

Therefore a 1D experiment should not be judged by a large change in the raw asymptotic exponent. The better diagnostic is Fourier recovery during training:

$$R_r(k,t) = \frac{|\widehat{f}_{r,t}(k)|}{|\widehat{f}^{\star}(k)|}, \qquad \beta_r(t) = \frac{d\log R_r(k,t)}{d\log k}. \tag{19}$$

A flatter recovery slope β_r means that high modes are learned more evenly relative to low modes.

This explains why the comparison to the classical spectral-bias paper must be made carefully. The classical claim is a learning-order claim: low frequencies are fitted faster. Our CPWL result is a geometric explanation of how rank can modify the finite-resolution Fourier transfer that governs this learning order. In dimensions $d \geq 2$, rank can suppress high-codimension intersections and make the observed spectrum closer to the low-codimension, facet-supported directions where Fourier decay is slowest. In $d = 1$, the raw CPWL tail is pinned near k^{-4}, so the right comparison is mode-wise recovery rather than raw asymptotic decay.

5 Experiments

We keep the experimental evidence deliberately short and defer settings to Appendix C. The high-frequency MMNN sweeps use rank-10, width-512 networks on `chirp1d` and `sumcos` targets at frequencies $8, 16, 32$, depths 3 and 6, with both frozen and trainable random-feature modes. They confirm that low-rank models can learn oscillatory targets, while optimizer choice and feature freezing remain important: AdamW is strongest on several trainable-feature `chirp1d` runs, whereas Muon and HT-Muon are competitive on some frozen-feature `sumcos` settings.

The CPWL theory also dictates how the experiments should be read. In one dimension, a ReLU network is a linear spline, and for derivative jumps Δ_j at knots t_j,

$$\widehat{f}(k) = -\frac{1}{(ik)^2}\sum_j \Delta_j e^{-ikt_j} + O(|k|^{-3}), \qquad |\widehat{f}(k)|^2 \sim |k|^{-4}A(k). \tag{20}$$

Thus a 1D experiment should not be judged by a large change in the raw asymptotic exponent. The better diagnostic is Fourier recovery during training,

$$R_r(k,t) = \frac{|\widehat{f}_{r,t}(k)|}{|\widehat{f}^{\star}(k)|}, \qquad \beta_r(t) = \frac{d\log R_r(k,t)}{d\log k}. \tag{21}$$

A flatter recovery slope β_r means that high modes are learned more evenly relative to low modes. This is why the comparison with the spectral-bias result of Rahaman et al. [8] must be made carefully: their claim is a learning-order claim, while our CPWL result explains how rank can modify the finite-resolution Fourier transfer governing that learning order. In dimensions $d \geq 2$, rank can suppress high-codimension intersections and move observed spectra toward low-codimension, facet-supported directions. In $d = 1$, the raw CPWL tail is pinned near k^{-4}, so the right comparison is mode-wise recovery rather than raw asymptotic decay.

The spectral-bias experiment isolates the rank effect. In a width-2^{15} Fourier/kernel surrogate, ranks $1, \ldots, 50$ are compared with a full-rank endpoint. The best rank is intermediate: $r = 4$ obtains MSE 0.250 versus 0.461 for full rank on sparse Fourier mixtures, and 0.443 versus 0.654 on dense mixtures. More importantly for Proposition 4.1, the recovery slope flattens from -0.94 and -1.20 at full rank to about -0.29 at rank 4, and the useful log-log scaling coefficient saturates at the same rank. This is the experimental signature that adding rank beyond the first plateau no longer improves finite-resolution high-frequency transfer.

The rank sweep is interpreted through the finite-time decomposition

$$\mathcal{E}_r(t) \le Ar^{-2s}\, \frac{Bd_{\text{eff}}(r)}{n} \sum_{k\in\Omega} |\widehat{f}^\star(k)|^2 e^{-2t\lambda_r(k)}. \tag{22}$$

The first term says rank cannot be too small; the second measures effective dimension; the third measures finite-time spectral transfer. The useful rank is the first rank that avoids the approximation bottleneck while keeping the recovery scaling flat.

6 Conclusion

Low rank is not only a compression device. In the random-feature mean-field regime, it preserves global convergence because frozen features maintain dense span and low-rank channel mixing only changes constants. In ReLU and CPWL regimes, it also changes the spectral bias through path-space constraints and Fourier face moments. The resulting picture is conservative but actionable: prove convergence in the frozen-feature low-rank model, then choose rank by spectral recovery on the target class. This makes low-rank networks a principled tool for highly oscillatory functions rather than merely a smaller approximation of full-rank networks.

A Assumptions and Dense-Span Property

The convergence theorem uses the following standard ingredients. The activation is Lipschitz and non-polynomial; for the cleanest theorem one may use sigmoid, tanh, or leaky ReLU with derivative bounded away from zero on the active region. Inputs and random features are bounded or sub-Gaussian. The random-feature law has dense support in $\mathbb{R}^d$. The trainable weights have sub-Gaussian initialization. The loss derivative is bounded and Lipschitz, and the loss is identifiable from its first-order condition.

Theorem A.1 (Universal approximation automatically maintained)**.** *If* $\operatorname{supp}(L^0(C_1)) = \mathbb{R}^d$ *and* φ_1 *is non-polynomial, then the class*

$$\left\{\varphi_1\left(\langle L^0(c_1), \cdot\rangle\right) : c_1 \in \Omega_1\right\} \tag{23}$$

has dense span in $L^2(\mathbb{P}_X)$ *throughout training.*

Proof. The statement is a direct application of the classical non-polynomial random-feature density theorem. The feature parameters are frozen, so their support cannot collapse during training. Therefore the dense span property holds at initialization and remains true for all $t \ge 0$. □

B Proof Details for Global Convergence

Proof of Theorem 3.2. Let $W(t) \to W^\star$. Since the dynamics are gradient flow, the time derivatives vanish at the limit in the weak topology used for the trajectory. The top-layer equation therefore implies

$$\mathbb{E}\left[\partial_2\mathcal{L}(Y, \hat{y}(X; W^\star))\varphi_L(H_L(c_L; X, W^\star))\right] = 0 \tag{24}$$

for all feature indices in the support. Propagating the same stationarity through the lower layers and using the frozen dense feature class gives

$$\mathbb{E}\left[\partial_2\mathcal{L}(Y, \hat{y}(X; W^\star)) \mid X = x\right] = 0 \tag{25}$$

for almost every x. By loss identifiability, the conditional loss is zero almost everywhere. Since $\mathcal{L} \ge 0$, this implies $\mathcal{L}(W^\star) = 0$, hence $W^\star$ is a global minimizer. □

C Additional Experimental Details

The high-frequency sweeps use `chirp1d` targets of the form

$$\cos(f\pi x^2) - 0.8\cos((f/3)\pi x^2) \tag{26}$$

and `sumcos` targets

$$\frac{1}{\sqrt{f}}\sum_{h=1}^{f}\cos(2\pi hx). \tag{27}$$

The main sweep uses width 512, rank 10, batch size 1024, 20000 optimization steps, depths 3 and 6, and frequencies $8, 16, 32$. The low-rank spectral-bias sweep uses the CSV data in `refs/lr_spectral_workshop/figdata`; its main conclusion is the intermediate optimum $r = 4$ in Fourier recovery.

Table 1: Representative frequency-16, depth-6 results from the current high-frequency sweeps. Lower is better.

Target	Mode	Optimizer	Test MSE
chirp1d	trainWb	AdamW	$8.8 \cdot 10^{-6}$
chirp1d	fixWb	Muon	$3.1 \cdot 10^{-3}$
chirp1d	fixWb	HT-Muon	$3.9 \cdot 10^{-3}$
sumcos	fixWb	AdamW	$5.1 \cdot 10^{-4}$
sumcos	fixWb	Muon	$1.8 \cdot 10^{-3}$
sumcos	trainWb	Muon	$6.5 \cdot 10^{-3}$

Table 2: Fourier/kernel rank sweep at width 2^{15}. Rank 4 balances approximation and optimization better than both very small ranks and the full endpoint.

Task	Best r	Best MSE	Full MSE	Slopes
Sparse	4	0.250	0.461	$-0.29/-0.94$
Dense	4	0.443	0.654	$-0.29/-1.20$

References

[1] F. Bach. Breaking the curse of dimensionality with convex neural networks. *Journal of Machine Learning Research*, 2017.

[2] A. Barron. Universal approximation bounds for superpositions of a sigmoidal function. *IEEE Transactions on Information Theory*, 1993.

[3] W. Czarnecki, S. Osindero, M. Jaderberg, G. Swirszcz, and R. Pascanu. Sobolev training for neural networks. In *NeurIPS*, 2017.

[4] R. Diaz and S. Robins. Fourier transforms of polytopes and solid angle sums. In *Integer Points in Polyhedra*, 2016.

[5] A. Jacot, F. Gabriel, and C. Hongler. Neural tangent kernel: Convergence and generalization in neural networks. In *NeurIPS*, 2018.

[6] G. Montufar, R. Pascanu, K. Cho, and Y. Bengio. On the number of linear regions of deep neural networks. In *NeurIPS*, 2014.

[7] P.-M. Nguyen and H. T. Pham. A rigorous framework for the mean-field limit of multilayer neural networks. 2023.

[8] N. Rahaman, A. Baratin, D. Arpit, F. Draxler, M. Lin, F. Hamprecht, Y. Bengio, and A. Courville. On the spectral bias of neural networks. In *ICML*, 2019.

[9] T. Zaslavsky. Facing up to arrangements: face-count formulas for partitions of space by hyperplanes. *Memoirs of the American Mathematical Society*, 1975.

NeurIPS Paper Checklist

The checklist is designed to encourage best practices for responsible machine learning research, addressing issues of reproducibility, transparency, research ethics, and societal impact. Do not remove the checklist: **The papers not including the checklist will be desk rejected.** The checklist should follow the references and follow the (optional) supplemental material. The checklist does NOT count towards the page limit.

Please read the checklist guidelines carefully for information on how to answer these questions. For each question in the checklist:

- You should answer [Yes], [No], or [N/A].
- [N/A] means either that the question is Not Applicable for that particular paper or the relevant information is Not Available.
- Please provide a short (1–2 sentence) justification right after your answer (even for [N/A]).

The checklist answers are an integral part of your paper submission. They are visible to the reviewers, area chairs, senior area chairs, and ethics reviewers. You will also be asked to include it (after eventual revisions) with the final version of your paper, and its final version will be published with the paper.

The reviewers of your paper will be asked to use the checklist as one of the factors in their evaluation. While [Yes] is generally preferable to [No], it is perfectly acceptable to answer [No] provided a proper justification is given (e.g., error bars are not reported because it would be too computationally expensive" or "we were unable to find the license for the dataset we used"). In general, answering [No] or [N/A] is not grounds for rejection. While the questions are phrased in a binary way, we acknowledge that the true answer is often more nuanced, so please just use your best judgment and write a justification to elaborate. All supporting evidence can appear either in the main paper or the supplemental material, provided in appendix. If you answer [Yes] to a question, in the justification please point to the section(s) where related material for the question can be found.

IMPORTANT, please:

- **Delete this instruction block, but keep the section heading "NeurIPS Paper Checklist"**,
- **Keep the checklist subsection headings, questions/answers and guidelines below.**
- **Do not modify the questions and only use the provided macros for your answers**.

1. **Claims**

   Question: Do the main claims made in the abstract and introduction accurately reflect the paper's contributions and scope?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the abstract and introduction do not include the claims made in the paper.
   - The abstract and/or introduction should clearly state the claims made, including the contributions made in the paper and important assumptions and limitations. A [No] or [N/A] answer to this question will not be perceived well by the reviewers.
   - The claims made should match theoretical and experimental results, and reflect how much the results can be expected to generalize to other settings.
   - It is fine to include aspirational goals as motivation as long as it is clear that these goals are not attained by the paper.

2. **Limitations**

   Question: Does the paper discuss the limitations of the work performed by the authors?

   Answer: **[TODO]**

   Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper has no limitation while the answer [No] means that the paper has limitations, but those are not discussed in the paper.
- The authors are encouraged to create a separate "Limitations" section in their paper.
- The paper should point out any strong assumptions and how robust the results are to violations of these assumptions (e.g., independence assumptions, noiseless settings, model well-specification, asymptotic approximations only holding locally). The authors should reflect on how these assumptions might be violated in practice and what the implications would be.
- The authors should reflect on the scope of the claims made, e.g., if the approach was only tested on a few datasets or with a few runs. In general, empirical results often depend on implicit assumptions, which should be articulated.
- The authors should reflect on the factors that influence the performance of the approach. For example, a facial recognition algorithm may perform poorly when image resolution is low or images are taken in low lighting. Or a speech-to-text system might not be used reliably to provide closed captions for online lectures because it fails to handle technical jargon.
- The authors should discuss the computational efficiency of the proposed algorithms and how they scale with dataset size.
- If applicable, the authors should discuss possible limitations of their approach to address problems of privacy and fairness.
- While the authors might fear that complete honesty about limitations might be used by reviewers as grounds for rejection, a worse outcome might be that reviewers discover limitations that aren't acknowledged in the paper. The authors should use their best judgment and recognize that individual actions in favor of transparency play an important role in developing norms that preserve the integrity of the community. Reviewers will be specifically instructed to not penalize honesty concerning limitations.

3. **Theory assumptions and proofs**

   Question: For each theoretical result, does the paper provide the full set of assumptions and a complete (and correct) proof?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include theoretical results.
   - All the theorems, formulas, and proofs in the paper should be numbered and cross-referenced.
   - All assumptions should be clearly stated or referenced in the statement of any theorems.
   - The proofs can either appear in the main paper or the supplemental material, but if they appear in the supplemental material, the authors are encouraged to provide a short proof sketch to provide intuition.
   - Inversely, any informal proof provided in the core of the paper should be complemented by formal proofs provided in appendix or supplemental material.
   - Theorems and Lemmas that the proof relies upon should be properly referenced.

4. **Experimental result reproducibility**

   Question: Does the paper fully disclose all the information needed to reproduce the main experimental results of the paper to the extent that it affects the main claims and/or conclusions of the paper (regardless of whether the code and data are provided or not)?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.

- If the paper includes experiments, a [No] answer to this question will not be perceived well by the reviewers: Making the paper reproducible is important, regardless of whether the code and data are provided or not.
- If the contribution is a dataset and/or model, the authors should describe the steps taken to make their results reproducible or verifiable.
- Depending on the contribution, reproducibility can be accomplished in various ways. For example, if the contribution is a novel architecture, describing the architecture fully might suffice, or if the contribution is a specific model and empirical evaluation, it may be necessary to either make it possible for others to replicate the model with the same dataset, or provide access to the model. In general. releasing code and data is often one good way to accomplish this, but reproducibility can also be provided via detailed instructions for how to replicate the results, access to a hosted model (e.g., in the case of a large language model), releasing of a model checkpoint, or other means that are appropriate to the research performed.
- While NeurIPS does not require releasing code, the conference does require all submissions to provide some reasonable avenue for reproducibility, which may depend on the nature of the contribution. For example
  (a) If the contribution is primarily a new algorithm, the paper should make it clear how to reproduce that algorithm.
  (b) If the contribution is primarily a new model architecture, the paper should describe the architecture clearly and fully.
  (c) If the contribution is a new model (e.g., a large language model), then there should either be a way to access this model for reproducing the results or a way to reproduce the model (e.g., with an open-source dataset or instructions for how to construct the dataset).
  (d) We recognize that reproducibility may be tricky in some cases, in which case authors are welcome to describe the particular way they provide for reproducibility. In the case of closed-source models, it may be that access to the model is limited in some way (e.g., to registered users), but it should be possible for other researchers to have some path to reproducing or verifying the results.

5. **Open access to data and code**

   Question: Does the paper provide open access to the data and code, with sufficient instructions to faithfully reproduce the main experimental results, as described in supplemental material?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that paper does not include experiments requiring code.
   - Please see the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - While we encourage the release of code and data, we understand that this might not be possible, so [No] is an acceptable answer. Papers cannot be rejected simply for not including code, unless this is central to the contribution (e.g., for a new open-source benchmark).
   - The instructions should contain the exact command and environment needed to run to reproduce the results. See the NeurIPS code and data submission guidelines (`https://neurips.cc/public/guides/CodeSubmissionPolicy`) for more details.
   - The authors should provide instructions on data access and preparation, including how to access the raw data, preprocessed data, intermediate data, and generated data, etc.
   - The authors should provide scripts to reproduce all experimental results for the new proposed method and baselines. If only a subset of experiments are reproducible, they should state which ones are omitted from the script and why.
   - At submission time, to preserve anonymity, the authors should release anonymized versions (if applicable).

- Providing as much information as possible in supplemental material (appended to the paper) is recommended, but including URLs to data and code is permitted.

6. **Experimental setting/details**

   Question: Does the paper specify all the training and test details (e.g., data splits, hyperparameters, how they were chosen, type of optimizer) necessary to understand the results?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The experimental setting should be presented in the core of the paper to a level of detail that is necessary to appreciate the results and make sense of them.
   - The full details can be provided either with the code, in appendix, or as supplemental material.

7. **Experiment statistical significance**

   Question: Does the paper report error bars suitably and correctly defined or other appropriate information about the statistical significance of the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The authors should answer [Yes] if the results are accompanied by error bars, confidence intervals, or statistical significance tests, at least for the experiments that support the main claims of the paper.
   - The factors of variability that the error bars are capturing should be clearly stated (for example, train/test split, initialization, random drawing of some parameter, or overall run with given experimental conditions).
   - The method for calculating the error bars should be explained (closed form formula, call to a library function, bootstrap, etc.)
   - The assumptions made should be given (e.g., Normally distributed errors).
   - It should be clear whether the error bar is the standard deviation or the standard error of the mean.
   - It is OK to report 1-sigma error bars, but one should state it. The authors should preferably report a 2-sigma error bar than state that they have a 96% CI, if the hypothesis of Normality of errors is not verified.
   - For asymmetric distributions, the authors should be careful not to show in tables or figures symmetric error bars that would yield results that are out of range (e.g., negative error rates).
   - If error bars are reported in tables or plots, the authors should explain in the text how they were calculated and reference the corresponding figures or tables in the text.

8. **Experiments compute resources**

   Question: For each experiment, does the paper provide sufficient information on the computer resources (type of compute workers, memory, time of execution) needed to reproduce the experiments?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the paper does not include experiments.
   - The paper should indicate the type of compute workers CPU or GPU, internal cluster, or cloud provider, including relevant memory and storage.
   - The paper should provide the amount of compute required for each of the individual experimental runs as well as estimate the total compute.

- The paper should disclose whether the full research project required more compute than the experiments reported in the paper (e.g., preliminary or failed experiments that didn't make it into the paper).

9. **Code of ethics**

   Question: Does the research conducted in the paper conform, in every respect, with the NeurIPS Code of Ethics `https://neurips.cc/public/EthicsGuidelines`?

   Answer: **[TODO]**

   Justification: **[TODO]**

   Guidelines:

   - The answer [N/A] means that the authors have not reviewed the NeurIPS Code of Ethics.
   - If the authors answer [No], they should explain the special circumstances that require a deviation from the Code of Ethics.
   - The authors should make sure to preserve anonymity (e.g., if there is a special consideration due to laws or regulations in their jurisdiction).

10. **Broader impacts**

    Question: Does the paper discuss both potential positive societal impacts and negative societal impacts of the work performed?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that there is no societal impact of the work performed.
    - If the authors answer [N/A] or [No], they should explain why their work has no societal impact or why the paper does not address societal impact.
    - Examples of negative societal impacts include potential malicious or unintended uses (e.g., disinformation, generating fake profiles, surveillance), fairness considerations (e.g., deployment of technologies that could make decisions that unfairly impact specific groups), privacy considerations, and security considerations.
    - The conference expects that many papers will be foundational research and not tied to particular applications, let alone deployments. However, if there is a direct path to any negative applications, the authors should point it out. For example, it is legitimate to point out that an improvement in the quality of generative models could be used to generate Deepfakes for disinformation. On the other hand, it is not needed to point out that a generic algorithm for optimizing neural networks could enable people to train models that generate Deepfakes faster.
    - The authors should consider possible harms that could arise when the technology is being used as intended and functioning correctly, harms that could arise when the technology is being used as intended but gives incorrect results, and harms following from (intentional or unintentional) misuse of the technology.
    - If there are negative societal impacts, the authors could also discuss possible mitigation strategies (e.g., gated release of models, providing defenses in addition to attacks, mechanisms for monitoring misuse, mechanisms to monitor how a system learns from feedback over time, improving the efficiency and accessibility of ML).

11. **Safeguards**

    Question: Does the paper describe safeguards that have been put in place for responsible release of data or models that have a high risk for misuse (e.g., pre-trained language models, image generators, or scraped datasets)?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper poses no such risks.

- Released models that have a high risk for misuse or dual-use should be released with necessary safeguards to allow for controlled use of the model, for example by requiring that users adhere to usage guidelines or restrictions to access the model or implementing safety filters.
- Datasets that have been scraped from the Internet could pose safety risks. The authors should describe how they avoided releasing unsafe images.
- We recognize that providing effective safeguards is challenging, and many papers do not require this, but we encourage authors to take this into account and make a best faith effort.

12. **Licenses for existing assets**

    Question: Are the creators or original owners of assets (e.g., code, data, models), used in the paper, properly credited and are the license and terms of use explicitly mentioned and properly respected?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not use existing assets.
    - The authors should cite the original paper that produced the code package or dataset.
    - The authors should state which version of the asset is used and, if possible, include a URL.
    - The name of the license (e.g., CC-BY 4.0) should be included for each asset.
    - For scraped data from a particular source (e.g., website), the copyright and terms of service of that source should be provided.
    - If assets are released, the license, copyright information, and terms of use in the package should be provided. For popular datasets, `paperswithcode.com/datasets` has curated licenses for some datasets. Their licensing guide can help determine the license of a dataset.
    - For existing datasets that are re-packaged, both the original license and the license of the derived asset (if it has changed) should be provided.
    - If this information is not available online, the authors are encouraged to reach out to the asset's creators.

13. **New assets**

    Question: Are new assets introduced in the paper well documented and is the documentation provided alongside the assets?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not release new assets.
    - Researchers should communicate the details of the dataset/code/model as part of their submissions via structured templates. This includes details about training, license, limitations, etc.
    - The paper should discuss whether and how consent was obtained from people whose asset is used.
    - At submission time, remember to anonymize your assets (if applicable). You can either create an anonymized URL or include an anonymized zip file.

14. **Crowdsourcing and research with human subjects**

    Question: For crowdsourcing experiments and research with human subjects, does the paper include the full text of instructions given to participants and screenshots, if applicable, as well as details about compensation (if any)?

    Answer: **[TODO]**

    Justification: **[TODO]**

Guidelines:

- The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
- Including this information in the supplemental material is fine, but if the main contribution of the paper involves human subjects, then as much detail as possible should be included in the main paper.
- According to the NeurIPS Code of Ethics, workers involved in data collection, curation, or other labor should be paid at least the minimum wage in the country of the data collector.

15. **Institutional review board (IRB) approvals or equivalent for research with human subjects**

    Question: Does the paper describe potential risks incurred by study participants, whether such risks were disclosed to the subjects, and whether Institutional Review Board (IRB) approvals (or an equivalent approval/review based on the requirements of your country or institution) were obtained?

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the paper does not involve crowdsourcing nor research with human subjects.
    - Depending on the country in which research is conducted, IRB approval (or equivalent) may be required for any human subjects research. If you obtained IRB approval, you should clearly state this in the paper.
    - We recognize that the procedures for this may vary significantly between institutions and locations, and we expect authors to adhere to the NeurIPS Code of Ethics and the guidelines for their institution.
    - For initial submissions, do not include any information that would break anonymity (if applicable), such as the institution conducting the review.

16. **Declaration of LLM usage**

    Question: Does the paper describe the usage of LLMs if it is an important, original, or non-standard component of the core methods in this research? Note that if the LLM is used only for writing, editing, or formatting purposes and does *not* impact the core methodology, scientific rigor, or originality of the research, declaration is not required.

    Answer: **[TODO]**

    Justification: **[TODO]**

    Guidelines:

    - The answer [N/A] means that the core method development in this research does not involve LLMs as any important, original, or non-standard components.
    - Please refer to our LLM policy in the NeurIPS handbook for what should or should not be described.